

EDUCATION

•Information Technology	2024-2028
AJAY KUMAR GARG ENGINEERING COLLEGE	CGPA: 7.4

PERSONAL PROJECTS

•Stock Price Prediction Using LSTM	GitHub Link
Python, LSTM, FastAPI, Time Series Forecasting	

- Built an LSTM-based deep learning model to forecast the next **60 trading days** using historical stock price data.
- Designed an end-to-end ML pipeline including data ingestion (Yahoo Finance), preprocessing, scaling, training, and evaluation.
- Implemented EMA-20 and EMA-50 technical indicators to generate automated **Golden and Death Cross** trading signals.
- Deployed the trained model as a **FastAPI REST service** enabling real-time inference and scalable access.

YouTube Video Q&A Chatbot using RAG	GitHub Link
LangChain, Hugging Face (Mistral), FAISS, Groq, NLP	

- Developed a Retrieval-Augmented Generation (RAG) chatbot to answer user queries using contextual information from YouTube video transcripts.
- Implemented the complete RAG pipeline including transcript extraction, text chunking, embedding generation, FAISS vector indexing, and response generation.
- Integrated conversational memory to maintain multi-turn context and improve response relevance and coherence.
- Leveraged the Groq LLM API to achieve **low-latency inference** compared to standard hosted LLM solutions.

TECHNICAL SKILLS AND INTERESTS

Programming Languages: C, Python

Data Science & ML Tools: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, PyTorch, Optuna

LLM & AI Frameworks: LangChain, LangGraph, Hugging Face, RAG Pipelines, Prompt Engineering

Backend & API Development: FastAPI, REST APIs,Streamlit

ML & AI Concepts: Supervised and Unsupervised Learning, Feature Engineering, ANN, CNN, RNN, LSTM, Transformers

Databases: SQL Server

Relevant Coursework: Data Structures and Algorithms, Operating Systems, Object-Oriented Programming, Database Management Systems

Areas of Interest: Software Engineering, Artificial Intelligence, Data Science, LLM-based Applications

ACHIEVEMENTS AND CERTIFICATES

- Earned **3-Star SQL Certification** on HackerRank, demonstrating strong proficiency in database querying and data manipulation
- Active **Machine Learning Member** at **GDG AKGEC**, contributing to ML discussions, workshops, and hands-on projects
- Solved **100+ Data Structures and Algorithms problems** on LeetCode, strengthening problem-solving and algorithmic thinking

PROFILE LINKS

•GFG •LeetCode